YUHE (MAGGIE) DENG
(716) 579-1833 | maggie.deng@sloan.mit.edu

EDUCATION

MIT SLOAN SCHOOL OF MANAGEMENT Cambridge, MA
Candidate for Master of Finance, February 2028 2026 - Present

- Intended concentration in Financial Engineering
- Anticipated Courses: Financial Engineering, Options and Futures Markets, Deep Learning, AI and Money, Algorithm
- Clubs: Quantitative Finance Club, Data Analytics Club

BRANDEIS UNIVERSITY Waltham, MA
B.S. in Applied Mathematics and B.S. in Quantitative Economics, summa cum laude 2022 - 2026

- GPA: 4.0/4.0, Dean's List All Semesters
- Relevant Coursework: Linear Algebra, Differential Equations, Probability and Statistics, Real Analysis, Stochastic Processes, Numerical Methods, Microeconomic and Macroeconomic Theory, Econometrics
- Honors: Arnold Shapiro Prize in Mathematics, Phi Beta Kappa Honors Society

EXPERIENCE

EMPIRICAL PANDEMIC UNEMPLOYMENT INSURANCE THESIS Waltham, MA
Independent Researcher 2025 - 2026

- Estimated causal labor-market effects of pandemic UI termination across 3,235 U.S. counties using a contiguous-border county-pair difference-in-differences design, finding modest unemployment declines but no statistically robust gains in employment, payroll, or establishment activity
- Constructed county-level panels from BLS and Census data in Python, forming 502 contiguous border county pairs (492 counties), with an alternative 907-pair sample constructed for robustness testing
- Leveraged event-study models with county-pair and time fixed effects, clustered standard errors, and pre-treatment trend tests to isolate causal effects from confounding regional trends

LAKEFRONT ASSET MANAGEMENT Beijing, China
Commodity Research Intern Summer 2025

- Developed a bottom-up industrial silicon cost model across 4 regions and 5 leading producers, decomposing raw material, electricity, labor expenses, and regional production structures to estimate producer-level cost curves and margins
- Stress-tested producer margins under changes in electricity costs, raw-material prices and market prices, identifying manufacturers and regions most exposed to commodity-price shocks
- Enhanced commodity price analysis with quantitative forecasting by building a time-series pipeline using two-year futures observations, estimating ARIMA specifications, and evaluating short-horizon predictions with out-of-sample MAPE <15%
- Presented a research report to the investment team, integrating cost-curve estimates, price forecasts, supply-demand conditions, regional production trends, and policy effects to support market assessment and investment discussions

SEGREGATION DYNAMICS COMPUTATIONAL MODELING RESEARCH Waltham, MA
Project Lead Spring 2025

- Modeled the emergence of residential segregation from individual location preferences using the Schelling framework, in which agents relocate when the share of similar neighbors falls below a specified tolerance threshold
- Engineered a Python agent-based simulation incorporating heterogeneous tolerance thresholds, local neighborhood interactions, and stochastic relocation, tracking convergence time and neighborhood sameness over 20,000-iteration runs
- Classified 4 distinct dynamic regimes by convergence behavior and population asymmetry, using heat maps and log-scaled convergence-time plots across population-tolerance combinations
- Directed experimental design, simulation validation, and results interpretation as project lead, coordinating a team through final analysis and presentation

BANK OF COMMUNICATIONS Hong Kong
Global Banking Intern Summer 2024

- Collaborated across compliance teams to screen 10,000+ cross-border transactions for state-owned enterprise subsidiaries, flagging compliance exceptions during account onboarding under regulatory deadlines
- Partnered with credit teams to assess corporate credit risk, analyzing financial statements and updating 5+ leverage, liquidity, and repayment-capacity metrics across two lending reports

ADDITIONAL INFORMATION

- Technical: Python, R, Stata, MATLAB, LaTeX
- Interests: skiing, yoga, Peking opera, calligraphy